JCTC Journal of Chemical Theory and Computation

pubs.acs.org/JCTC Article

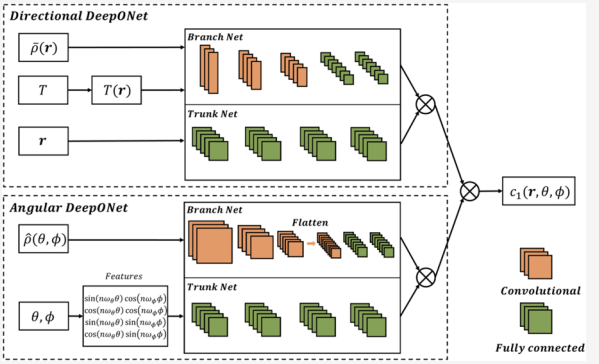
High-Dimensional Operator Learning for Molecular Density Functional Theory

Jinni Yang, Runtong Pan, Jikai Sun, and Jianzhong Wu*

Cite This: J. Chem. Theory Comput. 2025, 21, 5905−5915 Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

ABSTRACT: Classical density functional theory (cDFT) provides a systematic framework to predict the structure and thermodynamic properties of chemical systems through molecular density profiles. Whereas the statistical-mechanical framework is theoretically rigorous, its applications are often constrained by challenges in formulating a reliable free-energy functional and the complexity of solving multi-dimensional integro-differential equations. In this work, we established a convolutional operator learning method that effectively separates the high-dimensional molecular density profile into lower-dimensional components, thereby exponentially reducing the vast input space. The operator learning network demonstrates exceptional learning capabilities, accurately mapping the relationship between the molecular density profile and its one-body direct correlation function for an atomistic polarizable model of carbon dioxide. The machine-learning procedure can be generalized to more complex molecular systems, offering high-precision operator-cDFT calculations at a low computational cost.



INTRODUCTION

Classical density functional theory (cDFT) is a versatile method in statistical mechanics to describe the structure and thermodynamic properties of many-body systems based on the one-body density profiles.[1−3] Like its electronic counterpart, the practical applications of cDFT are often constrained by approximations in formulating a reliable free-energy functional and by the numerical complexity in solving the density profiles through variational minimization. Although highly accurate functionals can be formulated by using the fundamental measure theory and perturbation expansions,[4−7] cDFT calculations are predominantly limited to relatively simple molecular systems, often represented by coarse-grained models with only one-dimensional density inhomogeneity. While recent advances in machine-learning methods offer new avenues for developing more accurate functionals and efficient numerical algorithms,[8−18] existing studies primarily focus on simple fluids as represented by the hard-sphere or Lennard-Jones models. Besides, the operator relationships between density and potential fields are represented by conventional deep learning methods for functional learning that require exponentially more data to generalize effectively than operator learning.[19] In this work, we introduce a convolutional operator learning network for mapping high-dimensional density profiles to their corresponding one-body direct correlation functions. By applying the operator learning method to carbon dioxide (CO_2) modeled as three polarizable atoms, we demonstrate that operator-cDFT has advantages over functional learning in terms of accuracy, flexibility, and scalability while also addressing the computational challenges associated with high-dimensional function mappings.

Numerous molecular models have been developed for carbon dioxide.[20−27] Among the most notable is the elementary physical model (EPM2) introduced by Harris and Yung.[20] EPM2 effectively captures the key properties of CO_2 by adjusting the parameters in the pair potential function. Subsequent advancements have focused on incorporating bond flexibility,[21] molecular polarization,[22] and other many-body interactions,[23] significantly improving the precision of CO_2 simulations. In recent years, *ab initio* molecular dynamics simulations and machine-learned potentials have also been used to predict CO_2 phase behavior and thermodynamic properties.[28] From the practical perspective, as one of the primary greenhouse gases, carbon dioxide plays a pivotal role in global warming.[29] Human activities, such as fossil fuel combustion and industrial emissions, have drastically increased atmospheric CO_2 concentrations, contributing to rising temperatures and more frequent extreme weather events.[30,31] Reducing CO_2 emissions has become central to global climate policies. In addition to carbon capture and storage (CCS)

Received: March 25, 2025
Revised: May 6, 2025
Accepted: May 28, 2025
Published: June 5, 2025



Downloaded from pubs.acs.org/jctcce/article-pdf/21/12/5905/4126165/ct5c00484.pdf by ZHENGZHOU UNIV user on 09 October 2026

© 2025 American Chemical Society

https://doi.org/10.1021/acs.jctc.5c00484
J. Chem. Theory Comput. 2025, 21, 5905−5915

technologies for mitigating net emissions, CO_2 holds significant industrial value as it has been widely utilized in chemical production and oil and gas extraction.[32−35] CO_2 can also serve as a raw material for producing fuels and materials, promoting the development of a carbon circular economy.[36,37] These applications underscore the importance of fundamental research in bridging the knowledge gap between the molecular properties of CO_2 and its thermodynamic behavior under diverse environments, as represented by an external potential. Toward that end, an understanding of the intrinsic properties of CO_2 is crucial for industrial innovation and sustainable development.

CLASSICAL DFT AND DATA PREPARATION

The basic concepts of classical density functional theory (cDFT) for molecular systems have been well established.[38] For an inhomogeneous system of CO_2 molecules specified by temperature T and chemical potential μ, the grand canonical potential is defined as

$$\Omega[\rho(\boldsymbol{R})] = F^{id}[\rho(\boldsymbol{R})] + F^{exc}[\rho(\boldsymbol{R})] + \int d\boldsymbol{R}\rho(\boldsymbol{R})[V^{ext}(\boldsymbol{R}) - \mu] \quad (1)$$

where $\boldsymbol{R}$ represents the molecular configuration, i.e., the atomic positions of a CO_2 molecule, $\rho(\boldsymbol{R})$ denotes the one-body molecular density profile, F^{id} and F^{exc} are the ideal and excess intrinsic Helmholtz energy functionals, respectively, and $V^{ext}(\boldsymbol{R})$ is the one-body external potential. The ideal part of the intrinsic Helmholtz energy is the same as that of an ideal CO_2 gas with the same one-body density profile[39]

$$F^{id} = k_B T \int d\boldsymbol{R}\rho(\boldsymbol{R})\{\ln[\rho(\boldsymbol{R})q_I\Lambda^3] - 1\} \quad (2)$$

where $\Lambda = \sqrt{h^2/2\pi m k_B T}$ is the thermal wavelength, h is the Planck constant, m is the molecular mass, k_B is the Boltzmann constant, and q_I represents the single-molecule partition function. In this work, CO_2 is modeled as a rigid linear molecule, i.e., the molecular configuration is represented by the center of mass position $\boldsymbol{r}$ plus polar and azimuthal angles (θ,ϕ). Accordingly, we have $\boldsymbol{R} = (\boldsymbol{r},\theta,\phi)$ and $q_I = 4\pi$.

The excess part of the intrinsic Helmholtz energy accounts for intermolecular interactions and correlation effects. While this term is generally unknown for nonideal systems, its functional derivative with respect to the one-body density profile can be calculated from molecular dynamics or Monte Carlo simulations based on the Euler−Lagrange equation

$$c_1(\boldsymbol{R}) \equiv -\frac{\delta\beta F^{exc}}{\delta\rho(\boldsymbol{R})} = \ln[\rho(\boldsymbol{R})q_I\Lambda^3] + \beta[V^{ext}(\boldsymbol{R}) - \mu] \quad (3)$$

where $\beta = 1/k_B T$ and $c_1(\boldsymbol{R})$ is called the one-body direct correlation function. As demonstrated by Sammüller et al.,[11,18] the intrinsic relationship between $c_1(\boldsymbol{R})$ and $\rho(\boldsymbol{R})$ can be established with machine-learning methods. Selecting $c_1(\boldsymbol{R})$ as the target functional provides a crucial advantage due to its rapid spatial decay, which benefits both the training process and the prediction accuracy. If the objective were to learn the mapping relationship between $V^{ext}(\boldsymbol{R})$ and $\rho(\boldsymbol{R})$, the long-range nature of density correlations induced by the external potential would heavily impact the training process. From the operator relationship between $c_1(\boldsymbol{R})$ and $\rho(\boldsymbol{R})$, we can calculate the excess intrinsic Helmholtz energy using functional line integration

$$\beta F^{exc} = -\int_0^1 d\lambda \int d\boldsymbol{R}\rho(\boldsymbol{R})c_1(\boldsymbol{R};\ \lambda\rho(\boldsymbol{R})) \quad (4)$$

While conventional cDFT calculations rely on F^{exc} derived from various theoretical approximations, operator-cDFT employs machine-learning methods to establish the relationship between functions $c_1(\boldsymbol{R})$ and $\rho(\boldsymbol{R})$. Toward that end, we need to carry out molecular simulations to obtain high quality data for training the operator relationships. Although the data-driven approach offers little advantages over direct simulation to predict the thermodynamic properties of a specific system, cDFT allows for the applications of the intrinsic relationship to the same thermodynamic system under an arbitrary external potential. The universality of the *intrinsic* operator relationship makes it convenient for the systematic exploration of the thermodynamic behavior of CO_2 under diverse conditions.

The database for training the operator mapping of the one-body density profile to the one-body direct correlation function can be generated through grand canonical Monte Carlo (GCMC) simulations. In this work, CO_2 molecules are represented by the polarizable Gaussian-charge model proposed by Jiang et al.[22] As shown schematically in Figure 1, the atomic model assumes a rigid linear configuration and

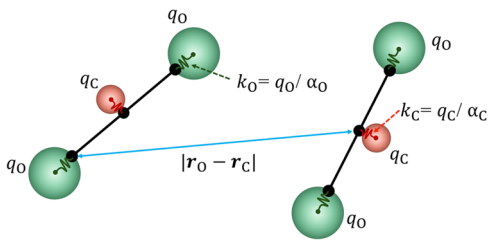


Figure 1. Schematic illustration of the polarizable Gaussian-charge model for the polarized CO_2.

adopts three classical Drude oscillators on the atomic sites to incorporate the polarization effects. While alternative models, either *ab initio* or coarse-grained, may also be used to generate the data, it has been demonstrated that the polarizable Gaussian-charge model is highly accurate for describing both the intermolecular potential and thermodynamic properties of CO_2 over diverse thermodynamic conditions.[22] In the Supporting Information, Figure S1 shows a comparison of various molecular models of CO_2 for predicting the second virial coefficient over a broad range of temperatures.

According to the polarizable Gaussian-charge model, the potential energy due to the interaction among CO_2 molecules can be represented by a pairwise additive potential

$$u(\boldsymbol{r}_i, \boldsymbol{r}_j) = \frac{q_i q_j}{4\pi\epsilon_0|\boldsymbol{r}_i - \boldsymbol{r}_j|}\mathrm{erf}\left(\frac{|\boldsymbol{r}_i - \boldsymbol{r}_j|}{\sqrt{2(\sigma_i^2 + \sigma_j^2)}}\right) + A_{ij} \exp(-B_{ij}|\boldsymbol{r}_i - \boldsymbol{r}_j|) - C_{ij}/|\boldsymbol{r}_i - \boldsymbol{r}_j|^6 \quad (5)$$

where q_i and q_j are the Gaussian (partial) charges of atoms i and j,[40] $|\boldsymbol{r}_i - \boldsymbol{r}_j|$ denotes the distance between their charge centers, and σ_i and σ_j are the widths of the charge distributions of the respective atoms. As shown in Figure 1, for each atom, its partial charge is attached to a fixed point on the molecular frame by a harmonic spring with the spring constant inversely proportional to the atomic polarizability α_i. In eq 5, parameters A_{ij}, B_{ij}, and C_{ij} are atom-specific and have the same meanings as

Downloaded from pubs.acs.org/jctce/article-pdf/21/12/5905/4126165/ct5c00484.pdf by ZHENGZHOU UNIV user on 09 October 2026

https://doi.org/10.1021/acs.jctc.5c00484
J. Chem. Theory Comput. 2025, 21, 5905−5915

Table 1. Model Parameters Describing the Molecular Structure, Atomic Polarization, and Site−Site Interactions (Adapted from Ref 22)

atom	A (MJ/mol)	B (Å^{-1})	C (kJ/mol/Å)	q (e)	σ (Å)	α (Å^3)	geometry
C	78.29	3.55	728	0.660	0.500	0.836	r_{CO} = 1.162 Å
O	329.71	3.93	1548	−0.330	0.900	0.836	∠OCO = 180°

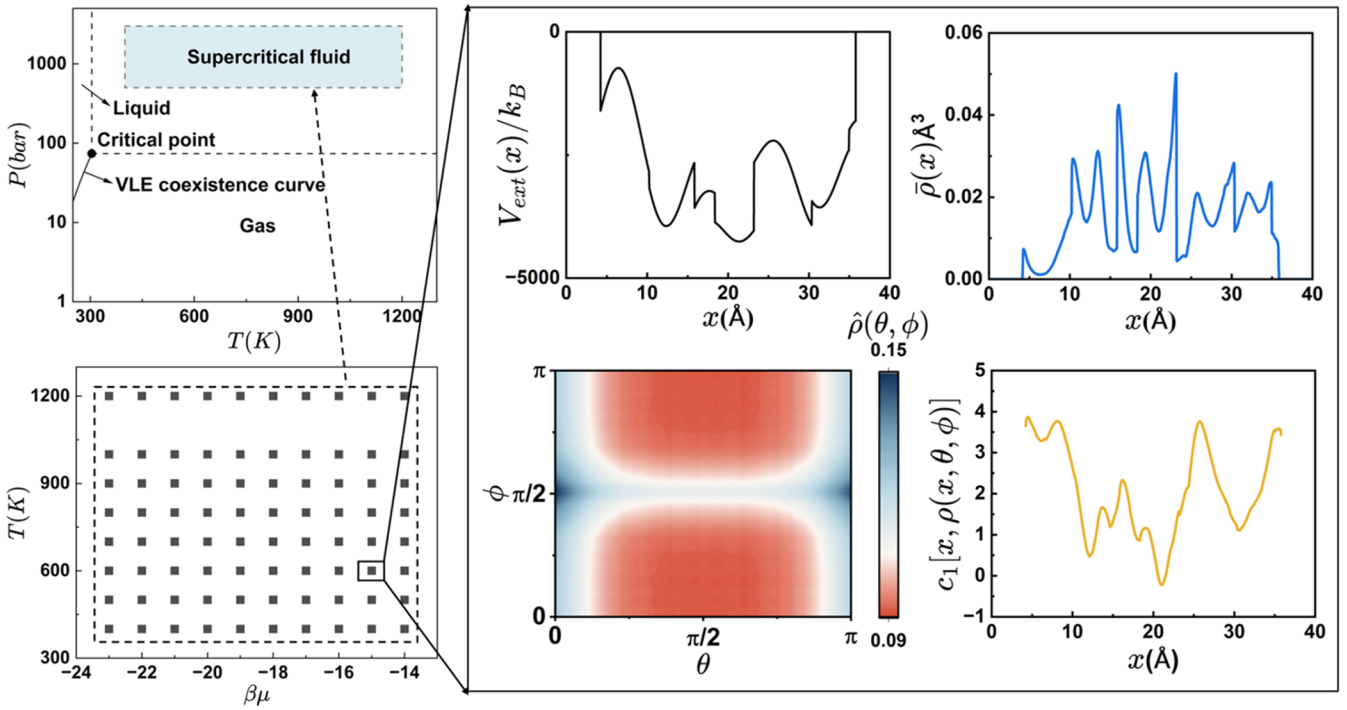


Figure 2. Training data obtained from GCMC simulation. The diagram on the left illustrates the simulation parameter space and the corresponding phase diagram. The plots on the right highlight one specific set of data obtained from GCMC simulation.

those that appeared in the Buckingham exp-6 potential.[41] Table 1 presents these molecular parameters, including those related to atomic charge, polarization, and molecular geometry. The cross-interaction parameters A_{ij} and B_{ij} are predicted from the Kong−Chakrabarty combining rules[42]

$$A_{ij} = \frac{1}{2}\left[A_{ii}\left(\frac{A_{ii}B_{ii}}{A_{jj}B_{jj}}\right)^{-B_{ii}/(B_{ii}+B_{jj})} + A_{jj}\left(\frac{A_{jj}B_{jj}}{A_{ii}B_{ii}}\right)^{-B_{jj}/(B_{ii}+B_{jj})}\right] \tag{5a}$$

$$B_{ij} = \frac{2B_{ii}B_{jj}}{B_{ii}+B_{jj}} \tag{5b}$$

and C_{ij} is obtained from the geometric mixing rule, $C_{ij} = \sqrt{C_{ii}\,C_{jj}}$.

The training data for $c_1(\boldsymbol{R})$ and $\rho(\boldsymbol{R})$ were generated by carrying out a GCMC simulation for CO_2 at supercritical conditions. For each atomic site, the external potential varies only in a single direction, which is designated as x. Therefore, $\rho(\boldsymbol{R})$ and $c_1(\boldsymbol{R})$ can be expressed in terms of 3-dimensional functions, $\rho(x,\theta,\phi)$ and $c_1(x,\theta,\phi)$, respectively. The atomic external potential consists of multiple sets of sinusoidal functions with different frequencies and phases, plus an equal number of discontinuous linear functions varying in a single direction of the simulation space[11]

$$V^{ext}(x)/k_B = \sum_{n=1}^{4} a_n\, T\, \sin\left(\frac{2\pi n x}{L} + \phi_n\right) + V_n(x) \tag{6}$$

where a_n and ϕ_n are Fourier coefficients and phase angles, L is the length of the simulation box in the x direction, and the discontinuous linear functions have the form

$$V_n(x) = \begin{cases} V_{n1} + \dfrac{x - x_{n1}}{x_{n2} - x_{n1}}(V_{n2} - V_{n1}), & \text{if } x_{n1} < x < x_{n2} \\ 0, & \text{otherwise} \end{cases} \tag{7}$$

where x_{n1} and x_{n2} are uniformly chosen between 0 and L, V_{n1} and V_{n2} are drawn from a normal distribution with a mean value of $-T$ and a variance of T. The latter represents the absolute temperature, which will be specified below. In eq 6, $a_{n=1,2,3,4}$ are randomly selected from a normal distribution with a zero mean and a variance of 1, and $\phi_{n=1,2,3,4}$ are from a uniform distribution over the interval $(0, 2\pi)$.

The GCMC calculations were conducted over 80 thermodynamic conditions for the bulk gas at supercritical conditions (Figure 2). In all cases, the simulation box has dimensions of 40 Å × 40 Å × 40 Å, with the periodic boundary conditions applied in all spatial directions. To avoid potential phase transitions, we take supercritical temperatures, T = 400, 500, 600, 700, 800, 900, 1000, and 1200 K, and a range of reduced chemical potentials, $\beta\mu$ = −{14,15,16,17,18,19,20,21,22,23}. In addition, we set V^{ext} =

Downloaded from pubs.acs.org/jctce/article-pdf/21/12/5905/4126165/ct5c00484.pdf by ZHENGZHOU UNIV user on 09 October 2026

https://doi.org/10.1021/acs.jctc.5c00484
J. Chem. Theory Comput. 2025, 21, 5905−5915

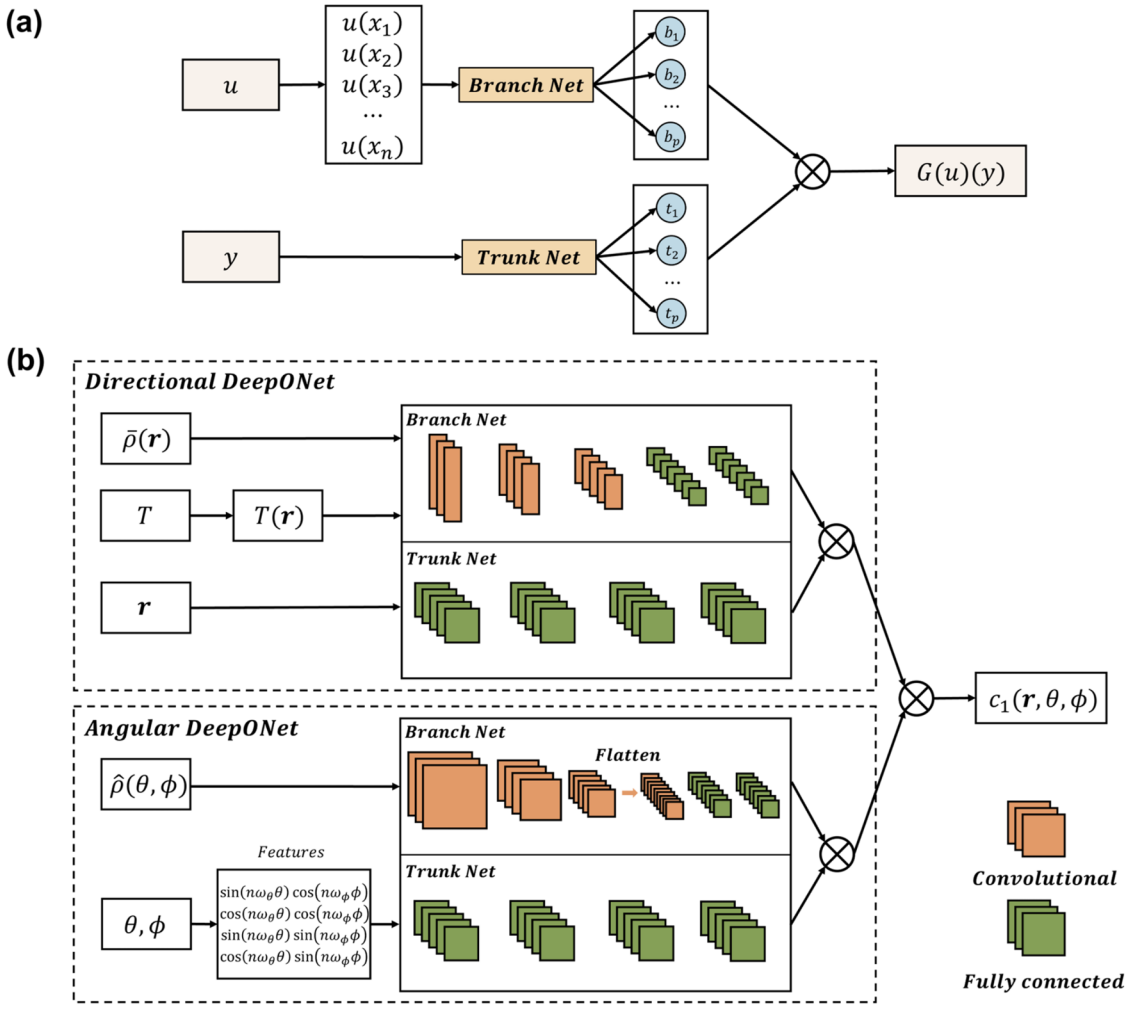


Figure 3. Illustration of the deep operator learning networks. (a) The general architecture of the "vanilla" DeepONet. (b) The convolutional operator learning networks (OLN) proposed in this work.

∞ for $|L/2-x|>|L/2-x_w|$, with x_w randomly selected from the interval of 2 to 6 Å, to effectively create a hard wall at the edge of the simulation domain. Additional details of the GCMC simulation are described in the Supporting Information (SI).

In operator-cDFT, the primary purpose of the molecular simulation is to obtain the training data for the molecular density profiles, $\rho(x,\theta,\phi)$, under different external potentials. To reduce the dimensionality of the molecular density profile, we have also sampled the angle-averaged local density $\bar{\rho}(\boldsymbol{r})$ and the position-averaged angular function $\hat{\rho}(\theta,\phi)$. It is important to note that for $\hat{\rho}(\theta,\phi)$, we only need to calculate its value in the ϕ direction within the range $[0, \pi]$. This is because the density profile exhibits a periodicity of π in the ϕ direction, which is smaller than the intrinsic periodicity of angular coordinate systems.

OPERATOR LEARNING METHODS

The operator learning methods applied in this work were inspired by the deep operator network (DeepONet) model and aim to approximate nonlinear continuous operators.[43] The DeepONet architecture is grounded in a rigorous mathematical framework known as the universal operator approximation,[44] which asserts that a neural network with a single hidden layer can accurately approximate any nonlinear continuous operator. Figure 3(a) presents the general architecture of the "vanilla" DeepONet, which is a basic form without any additional modifications or enhancements. The machine-learning model consists of two subneural networks, referred to as the branch net and the trunk net. For learning an operator with input function $u(x)$ and output function $G(y,u(x))$, the branch net samples n scattered points at fixed locations for each input, denoted as $u(x_1),u(x_2),\cdots,u(x_n)$. Meanwhile, the trunk net takes a number of scattered points for y, the independent variable of the output function. The final output of the operator learning is obtained by a dot product of the outputs of these two subnetworks

$$G(y, u(x)) = \sum_{k=1}^{p} \underbrace{b_k\,(u(x_1), u(x_2), \ldots, u(x_n))}_{\text{branch}} \underbrace{t_k\,(y)}_{\text{trunk}} \tag{8}$$

where p represents the dimension of the output vectors from the branch and trunk nets.

Intuitively, the branch-trunk structure can be interpreted as a trunk network, where each weight in the final layer is parametrized by a separate branch network. When DeepONet is applied to multidimensional mappings, the input to the branch net must also be multidimensional. Even when the dimensionality of each input is relatively low, the combined input $u(x_a,x_b,x_c)$ can become computationally infeasible, where

https://doi.org/10.1021/acs.jctc.5c00484
J. Chem. Theory Comput. 2025, 21, 5905−5915

Downloaded from pubs.acs.org/jctce/article-pdf/21/12/5905/4126165/ct5c00484.pdf by ZHENGZHOU UNIV user on 09 October 2026

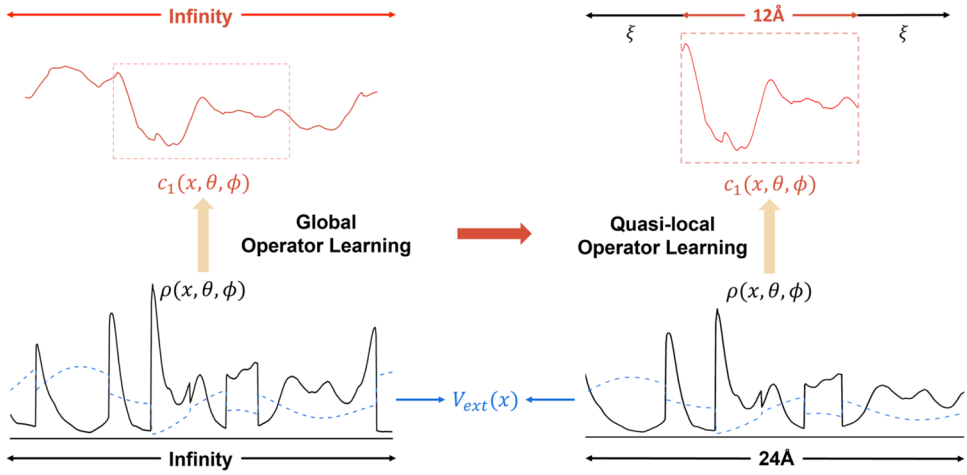


Figure 4. Illustration of the global and quasi-local operator learning schemes. The diagram on the left shows the global operator learning scheme, and the diagram on the right shows the quasi-local operator learning scheme.

subscripts a, b, and c denote the dimensions of each input. For a moderate three-dimensional (3D) problem with $a,b,c = 64$, even the latest NVIDIA Ampere GPUs lack sufficient memory to process a single training sample.[45] In the Supporting Information, we will demonstrate the specific structure of vanilla DeepONet when applied to solve multidimensional problems in classical DFT. Because of the computational challenges, we do not directly use the vanilla DeepONet to establish operator mapping from $\rho(\boldsymbol{R})$ to $c_1(\boldsymbol{R})$.

For systems consisting of linear molecules such as CO_2, both the molecular density profile and the one-body direct correlation function depend on the molecular position and orientation. In this case, we may expand $\rho(\boldsymbol{r},\theta,\phi)$ and $c_1(\boldsymbol{r},\theta,\phi)$ in terms of the spherical harmonics $Y_{ml}(\theta,\phi)$

$$\rho(\boldsymbol{r},\theta,\phi) = \sum_{l=0}^{\infty} \sum_{m=-l}^{l} g_{ml}(\boldsymbol{r}) Y_{ml}(\theta,\phi) \tag{9}$$

$$c_1(\boldsymbol{r},\theta,\phi) = \sum_{l=0}^{\infty} \sum_{m=-l}^{l} c_{ml}(\boldsymbol{r}) Y_{ml}(\theta,\phi) \tag{10}$$

where $g_{ml}(\boldsymbol{r})$ and $c_{ml}(\boldsymbol{r})$ are coefficients (viz, "projections") of the expansion. Instead of directly mapping the relationship between $\rho(\boldsymbol{r},\theta,\phi)$ and $c_1(\boldsymbol{r},\theta,\phi)$, $c_1(\boldsymbol{r},\theta,\phi)$ could be alternatively obtained from its spherical harmonic coefficients

$$c_1(\boldsymbol{r},\theta,\phi,\rho(\boldsymbol{r},\theta,\phi)) = \sum_{l=0}^{\infty} \sum_{m=-l}^{l} c_{ml}(\boldsymbol{r},\rho(\boldsymbol{r},\theta,\phi)) Y_{ml}(\theta,\phi) \tag{11}$$

Although eqs (9−11) are mathematically rigorous, they do not directly justify the decoupling of the angular and directional terms of the density profile. Nevertheless, eq 11 suggests that the projection functions of c_1, $c_{ml}(\boldsymbol{r},\rho(\boldsymbol{r},\theta,\phi))$ can be obtained through the convolution of position and angular variables using two DeepONets of lower dimensionality. As shown in Figure 3(b), one DeepONet uses the locally angle-averaged density profile $\overline{\rho}(\boldsymbol{r})$ as the input, and the other depends on the position-averaged angular function $\hat{\rho}(\theta,\phi)$. A major advantage of the convolutional operator learning network is that it enables the isolation of periodic angular components from position-dependent quantities. While $\overline{\rho}(\boldsymbol{r})$ generally does not exhibit periodic characteristics, we can apply the periodic boundary conditions specifically to $\hat{\rho}(\theta,\phi)$, thereby enhancing the training performance. In the Supporting Information, we provide additional details about the architecture and implementations of the directional DeepONet and the angular DeepONet. All codes and data supporting this work have been made publicly available on GitHub.[46]

As illustrated in Figure 4, we used a quasi-local operator learning scheme for training the functional relationship between $\rho(x,\theta,\phi)$ and $c_1(x,\theta,\phi)$. The strategy is different from the quasi-local scheme for simple fluids proposed by Sammüller et al.[11] While the latter is applied to the local value of c_1 with a segment of ρ as the input, the quasi-local operator learning treats c_1 as a function rather than discrete data points, preserving the operator relationship between c_1 and ρ. The quasi-local approximation assumes that the operator relationship between c_1 and ρ over the entire space can be approximated by that over a domain with a size larger than the correlation length. In our quasi-local operator learning scheme, the c_1 values from different sets of ρ are not independent but are uncorrelated with each other.

We established the quasi-local map between c_1 and ρ over a slit domain with a length of 12 Å in the perpendicular direction. The domain size is selected about twice the correlation length ($\xi \approx 6$ Å) at supercritical conditions, which is estimated from the molecular size of carbon dioxide (approximately 2 to 3 times the molecular diameter). Further investigation indicates that extending the correlation length does not lead to significant improvement in the performance of the trained model, and a detailed justification of our choice of correlation length is demonstrated in the Supporting Information. To mitigate the discrepancies between the local and global mappings, we incorporate the density information within the correlation length ξ beyond the specified range. This extended range is termed the cutoff distance. The quasi-local mapping is justified because the correlation length reflects the effective range over which molecules in the system interact with each other. In addition to the central part of the density profile and its corresponding c_1, two supplementary segments of the density profile, each with a width of $\xi = 6$ Å, are included in the training process to further enhance the training performance. Consequently, the actual mapping is from a ρ segment of 24 Å to a c_1 segment of 12 Å in length.

Since the mapping relationship $\rho(x,\theta,\phi) \rightarrow c_1(x,\theta,\phi)$ exists only within a specific ensemble and our GCMC simulations span over a broad range of temperatures, incorporating temperature as an additional input variable into the network

Downloaded from pubs.acs.org/jctce/article-pdf/21/12/5905/4126165/ct5c00484.pdf by ZHENGZHOU UNIV user on 09 October 2026

https://doi.org/10.1021/acs.jctc.5c00484
J. Chem. Theory Comput. 2025, 21, 5905−5915

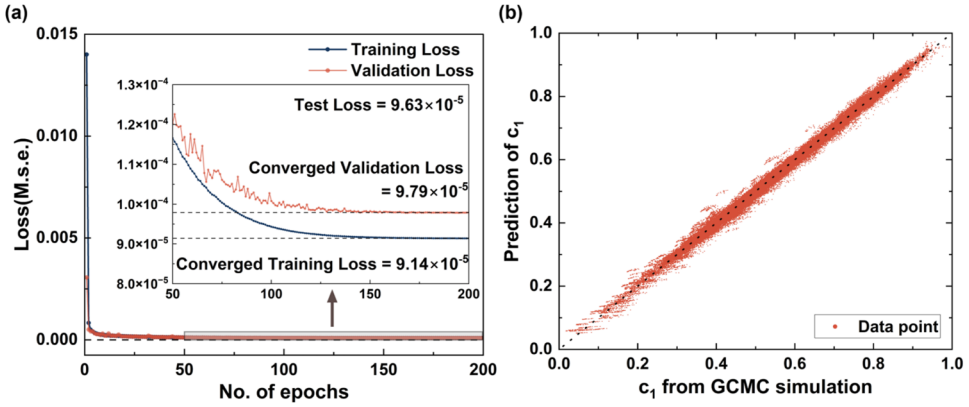


Figure 5. Illustration of the training result. (a) The evolution of training and validation loss over the number of epochs. (b) Comparison of the model prediction and data obtained from GCMC simulation.

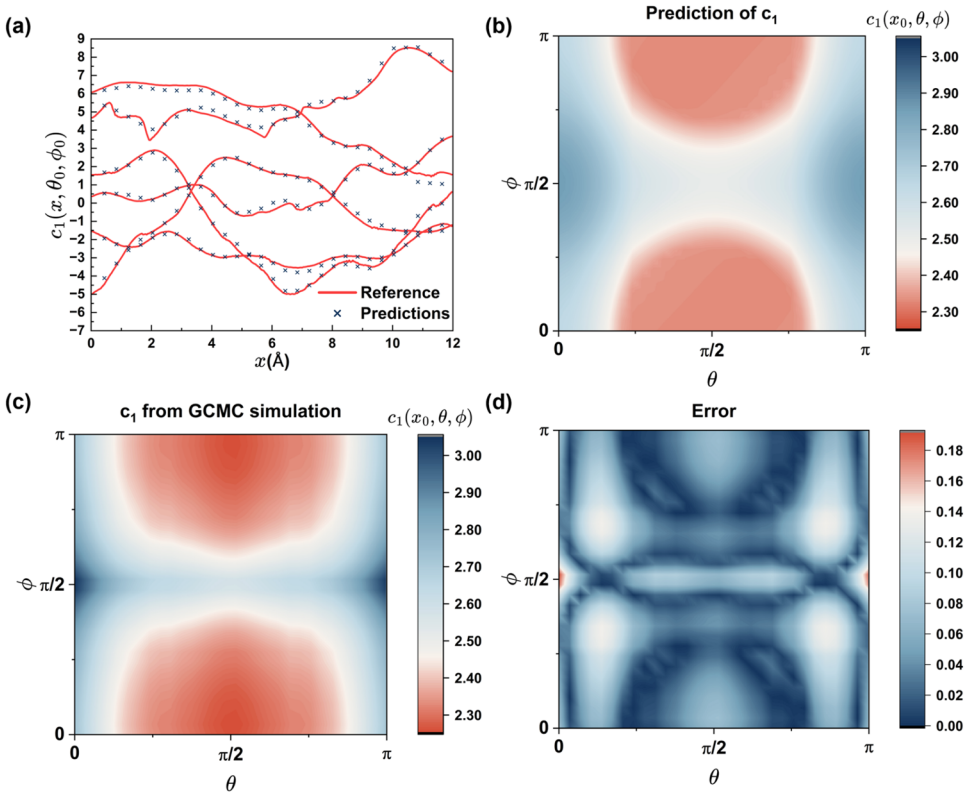


Figure 6. Predictions of one-body direct correlation functions. (a) The position component of $c_1(x,\theta,\phi)$ at 6 sets of fixed angular variables (θ_0,ϕ_0) randomly selected in the range of $[0, \pi]$. (b) The angular term (θ,ϕ) at fixed directional variable x_0 = 14.94 Å. (c) The reference of (b) obtained from GCMC. (d) The absolute error of the prediction on the angular plane.

is essential. To achieve this, we treat temperature as a constant function of x, denoted as $T(x)$, and incorporate it into training by adding an extra channel in the directional DeepONet, as shown in Figure 3(b). This approach has been demonstrated effective in incorporating the bulk conditions.[47−49]

In training our machine-learning models, each data point is represented as a sextuplet, denoted as $(\overline{\rho}(x), T(x), x, \hat{\rho}(\theta,\phi), (\theta,\phi), c_1(x,\theta,\phi))$, where $\overline{\rho}(x)$ is obtained by averaging $\rho(x,\theta,\phi)$ over all orientations at position x. $\overline{\rho}(x)$ is sampled within a range of 24 Å, starting from a random point in one set of density profiles. $T(x)$ is a constant function derived by repeating the chosen temperature across the directional sampling range. The exact value of x is randomly selected from the directional sampling range. Meanwhile, $\hat{\rho}(\theta,\phi)$ is obtained by using the same averaging method but within the angular slice under consideration and sampled on an equal-spaced grid of (π, π) with a $\pi/30$ gap in each dimension. Additionally, the value of (θ,ϕ) is randomly chosen from the angular sampling range, and $c_1(x,\theta,\phi)$ is randomly sampled from a combined space of the directional and angular dimensions but with a shorter sampling range of 12 Å in the x direction. Reducing the sampling range is necessary because mapping $\rho(x,\theta,\phi) \rightarrow c_1(x,\theta,\phi)$ is inherently global, making direct numerical handling impractical. Therefore, a well-designed training method is needed to overcome the challenges associated with this global relationship.

It should be noted that $c_1(x,\theta,\phi)$ cannot be directly determined from molecular simulation when the local density

https://doi.org/10.1021/acs.jctc.5c00484
J. Chem. Theory Comput. 2025, 21, 5905−5915

Downloaded from pubs.acs.org/jctce/article-pdf/21/12/5905/4126165/ct5c00484.pdf by ZHENGZHOU UNIV user on 09 October 2026

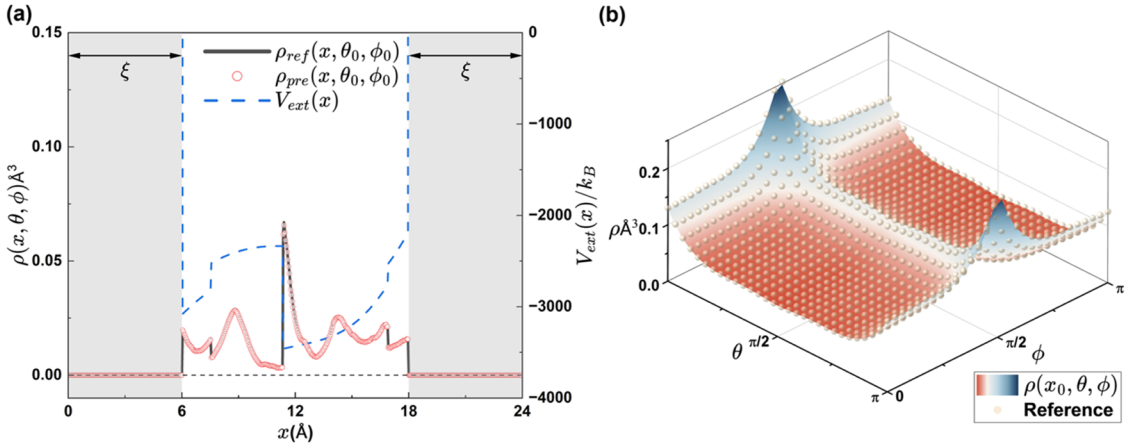


Figure 7. Predictions of the density profile from the data-enabled molecular cDFT. (a) The predicted density $\rho(x,\theta,\phi)$ with angular variables fixed and (b) the predicted density $\rho(x,\theta,\phi)$ with directional variables fixed. The temperature and reduced chemical potential of the system are randomly chosen as 400 K and −16, respectively.

vanishes, $\rho(x,\theta,\phi) = 0$. As shown in eq 3, this occurs within the hard-wall regions, excluding the direct sampling of $c_1(x,\theta,\phi)$ at positions where the external potential diverges. However, the zero-density regions are necessary for $\rho(x,\theta,\phi)$ sampling as they enable the trained model to recognize the presence of hard walls established in the simulation. The inclusion of zero-density data enhances the accuracy of the predicted density profile, as discussed in the following section.

NUMERICAL RESULTS

Prior to training our machine-learning model, we followed a conventional practice to partition the total database into three subsets: the training set, validation set, and test set, comprising 80, 10, and 10% of the total database, respectively. The training set is utilized for parameter optimization of the machine-learning model, while the validation set is employed to compute validation errors after each epoch, enabling us to monitor the training process and assist in hyperparameter tuning. The test set, on the other hand, is excluded from the training process and is exclusively reserved for evaluating test errors after completion of training. The specific training procedures are provided in the Supporting Information for detailed review.

Figure 5 illustrates the standard metrics of the training process to establish the operator relationship between $\rho(x,\theta,\phi)$ and $c_1(x,\theta,\phi)$. Figure 5(a) shows the evolution of the training and validation losses over the number of epochs. The training loss stabilizes at 9.14×10^{-5}, while the validation loss converges slightly higher at 9.79×10^{-5}, matching the test error. The close agreement between the training and validation losses signifies the absence of overfitting, as their convergence suggests that the model generalizes well to unseen data without excessive reliance on the training set. The operator learning model prediction from the test set is presented in Figure 5(b), which shows high consistency between the prediction and reference, namely, the ground truth values of c_1 obtained from GCMC simulation. This indicates that the machine-learning model demonstrates strong generalizability to new patterns that were not included in the training set.

To test the performance of the operator learning network, we first utilized the trained neural-operator model to predict the direct correlation function over the entire spatial domain. Figure 6(a,b) shows the prediction of $c_1(x,\theta,\phi)$ with either the angular variables or the directional variable fixed, respectively. In Figure 6(a), predictions are performed under the thermodynamic conditions of T = {700, 900, 700, 1200, 400, 500} K and 6 sets of (θ_0,ϕ_0) are randomly selected in the range of $[0, \pi]$, i.e., $[(\theta_0,\phi_0) = (5\pi/6, 2\pi/5),(\pi/3, 3\pi/10),(2\pi/3, 2\pi/15),(23\pi/30, \pi/5),(4\pi/5, \pi/6),(3\pi/5, 2\pi/5)]$. In Figure 6(b), prediction is performed under the thermodynamic conditions of T = 600 K and x_0 = 14.94 Å is randomly selected from $x \in [0, 24]$ Å. While the position-averaged angular function exhibited notable discrepancy at certain ranges of molecular angles, the machine-learning model shows excellent overall predictive capabilities. In particular, the test loss consistently stayed within a narrow range of 1×10^{-4}, indicating robust performance. The excellent performance of our machine-learning model in predicting the one-body direct correlation function over the multidimensional continuous space indicates that it is not merely a local interpolation scheme based on the training data, but it has truly learned the mapping relationship between $\rho(x,\theta,\phi)$ and $c_1(x,\theta,\phi)$.

To further validate our neural-operator model, we have predicted $\rho(x,\theta,\phi)$ under different external potentials by solving the Euler−Lagrange equation

$$\rho(x, \theta, \phi) = \frac{1}{4\pi\Lambda^3} \exp[c_1(x, \theta, \phi, \overline{\rho}(x), \hat{\rho}(\theta, \phi)) + \beta(\mu - V_{ext}(x))] \quad (12)$$

Given an expression for the reduced one-body potential $\beta(\mu - V_{ext}(x))$ along with the operator relationship for $c_1(x,\theta,\phi)$, we can self-consistently determine the density profiles from eq 12 using the Picard iteration. Because of the long-range nature of density correlations induced by intermolecular interactions, it is crucial to implement the quasi-local approximation with sufficient width in predicting the molecular density profile. To account for the boundary effects, we choose ξ as the width of $c_1(x,\theta,\phi)$ within the hard wall, where the molecular density profile vanishes. The extended density profile effectively cuts off the influence of the unseen potential beyond the domain under consideration. The specific scheme is to set $V_{ext}(x) = \infty$ and $\rho(x,\theta,\phi) = 0$ for $|x - x_w| < \xi$, where x_w denotes the distance to the hard-wall boundary.

Downloaded from pubs.acs.org/jctce/article-pdf/21/12/5905/4126116/5c00484.pdf by ZHENGZHOU UNIV user on 09 October 2026

https://doi.org/10.1021/acs.jctc.5c00484
J. Chem. Theory Comput. 2025, 21, 5905−5915

As shown in Figure 7, the neural-operator model yields an accurate prediction of $\rho(x,\theta,\phi)$ under an arbitrarily selected external potential. Here, the angular variables are randomly fixed to $(\pi/5, 2\pi/3)$, and the directional variable is randomly fixed to 12 Å. In both cases, the density profiles predicted by cDFT over the entire spatial domains are highly accurate, with the overall average errors converging around 1×10^{-4} after L_2-normalization. However, the accuracy of the predictions decreases significantly when attempting to predict under subcritical conditions, where a phase-separated system with distinct liquid and gas domains is expected to emerge.[18] We believe that the trained model has likely learned the interpolation ability between the given temperatures but lacks sufficient extrapolation capability, which is considerably more challenging to achieve in practice.

We have further justified the functional relationship between $c_1(x,\theta,\phi)$ and $\rho(x,\theta,\phi)$ by comparing the projection functions obtained from the spherical harmonic expansion $\rho(x,\theta,\phi)$

$$g_{ml}(r) = \int_0^{2\pi}\int_0^{\pi} \rho(r,\theta,\phi)Y_{ml}^{*}(\theta,\phi)\sin\theta \mathrm{d}\theta\mathrm{d}\phi \tag{13}$$

The leading coefficients, with $m,l = \{(0,0),(1,0),(2,0)\}$, are given by

$$g_{00}(r) = \int_0^{2\pi}\int_0^{\pi} \frac{\rho(r,\theta,\phi)\sin\theta}{\sqrt{4\pi}} \mathrm{d}\theta\mathrm{d}\phi \tag{14}$$

$$g_{10}(r) = \int_0^{2\pi}\int_0^{\pi} \sqrt{\frac{3}{4\pi}}\rho(r,\theta,\phi)\cos\theta\ \sin\theta \mathrm{d}\theta\mathrm{d}\phi \tag{15}$$

$$g_{20}(r) = \int_0^{2\pi}\int_0^{\pi} \sqrt{\frac{5}{16\pi}}\rho(r,\theta,\phi)(3\cos^2\theta - 1) \sin\theta \mathrm{d}\theta\mathrm{d}\phi \tag{16}$$

Figure 8 compares the results predicted from cDFT using the Euler−Lagrange equation and those obtained from molecular simulations. We can see that the cDFT predictions are highly accurate, corroborating the results shown in Figure 7.

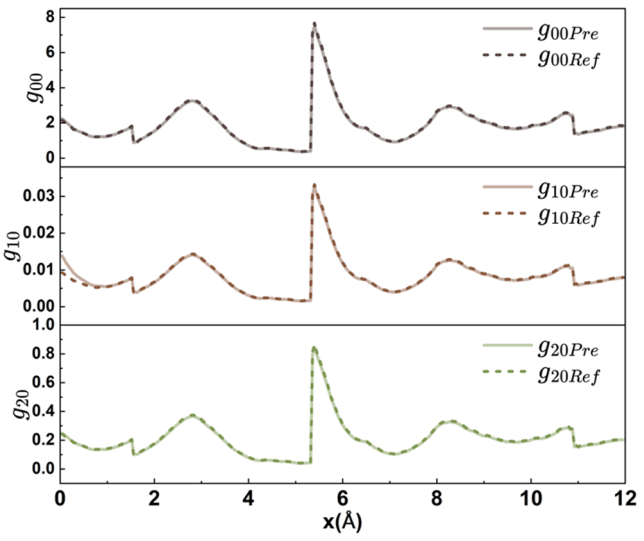


Figure 8. Comparison of the leading projection functions obtained from the spherical harmonic expansion of the molecular density profile. Subscript Pre means that $\rho(x,\theta,\phi)$ is predicted by cDFT, and ref denotes the results obtained from molecular simulation.

DISCUSSION

A conventional molecular simulation scales $O(n)$ for systems with truncated pairwise potentials and $O(nlog(n))$ for simulating systems with long-range interactions, where n is the number of atoms or molecules.[50] However, the machine-learning technique scales $O(1)$, implying that the computational cost is the same for either large or small systems, depending only on the length scale of intermolecular correlations. Therefore, compared with GCMC simulations, the neural-operator model can predict the equilibrium density profile with negligible computational cost. Importantly, the machine-learning model can be applied to the same thermodynamic system under any external potential.

In the development of the machine-learning models, we also considered the mappings from the projection functions of $\rho(\boldsymbol{r},\theta,\phi)$, $g_{ml}(\boldsymbol{r})$ to the projection functions of $c_1(\boldsymbol{r},\theta,\phi)$ and $c_{ml}(\boldsymbol{r})$. While the procedure is rigorous, it was not adopted in this work because of the following specific issue with the training. The mapping between $g_{ml}(\boldsymbol{r})$ and $c_{ml}(\boldsymbol{r})$ is not from real space to real space but from one Hilbert space to another Hilbert space, which leads to a dramatic increase in computational cost. It should be noted that the harmonic spherical expansion scheme $\rho_{\mathrm{ml}}(x) \leftrightarrow c_{ml}(x)$ will not reduce the dimensionality of training even though the angular variables are eliminated because all of the coefficients are required to fully represent the original functions. Even with the truncation of l in expansion, we still face a fundamental dilemma: computational tractability and acceptable accuracy cannot be simultaneously achieved. For example, with the optimistic assumption that the expansions for $\rho(\boldsymbol{r},\theta,\phi)$ and $c_1(\boldsymbol{r},\theta,\phi)$ converge on the first few spherical harmonics, we need at least six sets of basis functions for the mapping (with the maximum $l = 2$ in this case), which is not feasible for training sub-DeepONets used in this work.

The operator learning scheme proposed in this work can be naturally applied to other molecular systems. For other linear molecules, such as nitrogen, the density profile can also be expressed in terms of the molecular position and polar angles similar to that for CO_2. After obtaining a data set for $\rho(\boldsymbol{r},\theta,\phi)$ from GCMC or *ab initio* simulations, we can readily calculate the angle-averaged local density $\overline{\rho}(\boldsymbol{r})$ and the position-averaged angular function $\hat{\rho}(\theta,\phi)$ and establish the functional relationship $[\overline{\rho}(\boldsymbol{r}),\hat{\rho}(\theta,\phi)]\rightarrow c_1(\boldsymbol{r},\theta,\phi)$ using the operator learning network proposed in this work. Subsequently, we can predict the thermodynamic properties within the cDFT framework. The whole scheme is illustrated in Figure 9. For nonlinear molecules with a rigid configuration, three independent angular variables (θ,ϕ,γ) will be needed to fully represent their density profile. In that case, the density profile and one-body direct correlation function can be similarly separated in terms of the directional and angular terms. If the molecules are flexible, we must establish the operator relationships between c_1 and ρ for each atom (or interacting site), i.e., $c_{1,i} = c_{1,i}(\rho_1,\rho_2,\cdots,\rho_n)$, where $\rho_1,\rho_2,\cdots,\rho_n$ represent the density profile of each atom. Because the atomic density profiles are treated as parallel inputs, the operator learning network can be similarly used to establish functional relationships between c_1 and ρ for atomic sites.

In addition, the growing success of large language models has inspired analogous "large models" across scientific

Downloaded from pubs.acs.org/jctce/article-pdf/21/12/5905/4126165/ct5c00484.pdf by ZHENGZHOU UNIV user on 09 October 2026

https://doi.org/10.1021/acs.jctc.5c00484
J. Chem. Theory Comput. 2025, 21, 5905−5915

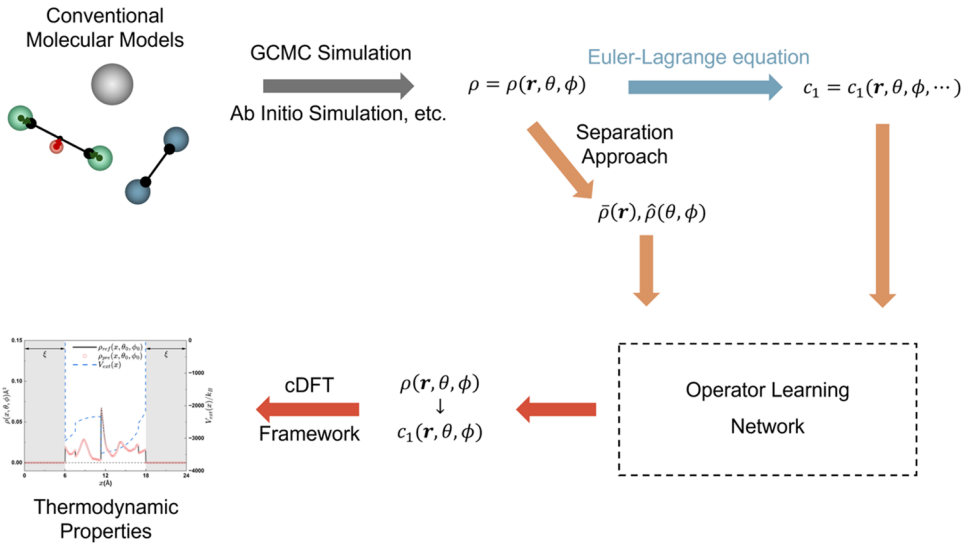


Figure 9. Illustration of a general operator learning scheme to establish the functional relationship between c_1 and ρ for a thermodynamic system consisting of linear molecules.

domains, demonstrating remarkable generalization capabilities.[51−54] Notable examples include the "large atomic models" that achieve transferability across diverse molecular systems.[53] Building on these advances, we envision the potential for a "universal model for neural density functionals" in the cDFT field. This would be a generalizable framework applicable to arbitrary molecular systems, as we can build corresponding models for unique molecular systems. While such a model remains future work, which requires extensive preliminary work and database support, we believe that the operator learning framework developed in this study could provide essential theoretical and computational foundations for its realization.

It should be noted that the extrapolation ability of machine-learning models remains poorly understood from a theoretical perspective, even within the machine-learning community. To further explore the transferability of our cDFT model, we speculate that the treatment of temperature as an input variable may be a critical factor. While representing temperature as a constant-value function in operator training has proven effective in certain applications, as shown in this work, this approach presents inherent contradictions: temperature is fundamentally a scalar parameter rather than a spatial function. By encoding it as a constant-value function without explicitly constraining its invariance, we may inadvertently introduce mathematical inconsistencies that limit model generalizability. Future studies, thus, should investigate more physically grounded approaches to incorporate thermodynamic variables, including temperature and chemical potential, while preserving their intrinsic properties as state parameters.

CONCLUSIONS

In this work, we proposed a convolutional operator learning network for mapping the functional relationship between the molecular density profile and the one-body direct correlation function of carbon dioxide within the framework of a polarizable atomic model. The machine-learning model leverages GCMC simulations and cDFT calculations to provide accurate and efficient predictions for complex molecular systems, enabling the exploration of diverse thermodynamic regimes. The computational efficiency makes the data-enabled cDFT a practical and scalable approach without compromising on precision. Given the broad applicability of dimension reduction techniques to diverse molecular systems, we expect that the operator learning scheme will unlock new avenues for a broader application of cDFT.

While our operator training method is robust when applied to supercritical conditions, it is foreseeable that near the critical point, the quasi-local approximation will no longer be reliable due to the divergence of the correlation length. In addition to addressing the breakdown of the quasi-local approximation near the critical point, enhancements in the accuracy of the databases used for operator learning could be explored in future work, with first-principles molecular simulations emerging as a promising avenue for further improvement. For practical applications, we could also concentrate on utilizing the operator learning method to predict the phase behavior and applications associated with CO_2 capture and conversion, thereby broadening the applicability beyond traditional simulation techniques.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jctc.5c00484.

Additional technical details of Grand Canonical Monte Carlo (GCMC) simulation; the second virial coefficient results of 4 different CO_2 models; the specific structure of vanilla DeepONet when applied to solve multidimensional problems in classical DFT; additional details of the revised operator learning network; the specific training procedure and metrics; and the justification regarding the choice of correlation length (PDF)

AUTHOR INFORMATION

Corresponding Author

Jianzhong Wu − *Department of Chemical and Environmental Engineering, University of California, Riverside, California 92521, United States;* orcid.org/0000-0002-4582-5941; Email: jwu@engr.ucr.edu

Downloaded from pubs.acs.org/jctce/article-pdf/21/12/5905/4126116/ct5c00484.pdf by ZHENGZHOU UNIV user on 09 October 2026

https://doi.org/10.1021/acs.jctc.5c00484
J. Chem. Theory Comput. 2025, 21, 5905−5915

Authors

Jinni Yang − *College of Physics, Jilin University, Changchun, Jilin 130015, P. R. China;* orcid.org/0009-0001-6907-6535

Runtong Pan − *Department of Chemical and Environmental Engineering, University of California, Riverside, California 92521, United States*

Jikai Sun − *Department of Chemical and Environmental Engineering, University of California, Riverside, California 92521, United States*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.jctc.5c00484

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This research is made possible through financial support from the NSF-DFG Lead Agency Activity in Chemistry and Transport in Confined Spaces under Grant No. NSF 2234013. The authors thank Miguel Liu-Schiaffini from Caltech for many fruitful discussions and suggestions from the operator learning side. J.Y. also thanks Wenhao Yu from Jilin University, Linyue Yang from Zhejiang University, and Lizhe Chen from Tsinghua University for helpful discussions regarding coding issues.

■ REFERENCES

(1) Evans, R. The nature of the liquid-vapour interface and other topics in the statistical mechanics of non-uniform, classical fluids. *Adv. Phys.* **1979**, *28* (2), 143−200.

(2) Chandler, D.; McCoy, J. D.; Singer, S. J. Density functional theory of nonuniform polyatomic systems. I. General formulation. *J. Chem. Phys.* **1986**, *85* (10), 5971−5976.

(3) Wu, J.; Li, Z. Density-functional theory for complex fluids. *Annu. Rev. Phys. Chem.* **2007**, *58*, 85−112.

(4) Bursik, B.; Eller, J.; Gross, J. Predicting Solvation Free Energies from the Minnesota Solvation Database Using Classical Density Functional Theory Based on the PC-SAFT Equation of State. *J. Phys. Chem. B* **2024**, *128* (15), 3677−3688.

(5) Valiev, M.; Chuev, G. N.; Fedotova, M. V. CDFTPY: A python package for performing classical density functional theory calculations for molecular liquids. *Comput. Phys. Commun.* **2022**, *276*, No. 108338.

(6) Borgis, D.; Luukkonen, S.; Belloni, L.; Jeanmairet, G. Accurate prediction of hydration free energies and solvation structures using molecular density functional theory with a simple bridge functional. *J. Chem. Phys.* **2021**, *155* (2), No. 024117.

(7) Liu, Y.; Fu, J.; Wu, J. Z. High-Throughput Prediction of the Hydration Free Energies of Small Molecules from a Classical Density Functional Theory. *J. Phys. Chem. Lett.* **2013**, *4* (21), 3687−3691.

(8) Simon, A.; Weimar, J.; Martius, G.; Oettel, M. Machine Learning of a Density Functional for Anisotropic Patchy Particles. *J. Chem. Theory Comput.* **2024**, *20* (3), 1062−1077.

(9) Sammüller, F.; Schmidt, M. Neural density functionals: Local learning and pair-correlation matching. *Phys. Rev. E* **2024**, *110* (3), No. L032601.

(10) Malpica-Morales, A.; Yatsyshin, P.; Duran-Olivencia, M. A.; Kalliadasis, S. Physics-informed Bayesian inference of external potentials in classical density-functional theory. *J. Chem. Phys.* **2023**, *159* (10), No. 104109.

(11) Sammüller, F.; Hermann, S.; de Las Heras, D.; Schmidt, M. Neural functional theory for inhomogeneous fluids: Fundamentals and applications. *Proc. Natl. Acad. Sci. U.S.A.* **2023**, *120* (50), No. e2312484120.

(12) Yatsyshin, P.; Kalliadasis, S.; Duncan, A. B. Physics-constrained Bayesian inference of state functions in classical density-functional theory. *J. Chem. Phys.* **2022**, *156* (7), No. 074105.

(13) Fang, X. Y.; Gu, M. Y.; Wu, J. Z. Reliable emulation of complex functionals by active learning with error control. *J. Chem. Phys.* **2022**, *157* (21), No. 214109.

(14) Cats, P.; Kuipers, S.; de Wind, S.; van Damme, R.; Coli, G. M.; Dijkstra, M.; van Roij, R. Machine-learning free-energy functionals using density profiles from simulations. *APL Mater.* **2021**, *9* (3), No. 031109.

(15) Lin, S. C.; Martius, G.; Oettel, M. Analytical classical density functionals from an equation learning network. *J. Chem. Phys.* **2020**, *152* (2), No. 021102.

(16) Lin, S.-C.; Oettel, M. A classical density functional from machine learning and a convolutional neural network. *SciPost Phys.* **2019**, *6* (2), No. 025.

(17) Santos-Silva, T.; Teixeira, P. I. C.; Anquetil-Deck, C.; Cleaver, D. J. Neural-network approach to modeling liquid crystals in complex confinement. *Phys. Rev. E* **2014**, *89* (5), No. 053316.

(18) Sammüller, F.; Schmidt, M.; Evans, R. Neural Density Functional Theory of Liquid-Gas Phase Coexistence. *Phys. Rev. X* **2025**, *15* (1), No. 011013.

(19) Wu, J. Z.; Gu, M. Y. Perfecting Liquid-State Theories with Machine Intelligence. *J. Phys. Chem. Lett.* **2023**, *14* (47), 10545−10552.

(20) Harris, J. G.; Yung, K. H. Carbon Dioxide's Liquid-Vapor Coexistence Curve and Critical Properties As Predicted by a Simple Molecular Model. *J. Phys. Chem. A* **1995**, *99*, 12021−12024.

(21) Cygan, R. T.; Romanov, V. N.; Myshakin, E. M. Molecular Simulation of Carbon Dioxide Capture by Montmorillonite Using an Accurate and Flexible Force Field. *J. Phys. Chem. C* **2012**, *116* (24), 13079−13091.

(22) Jiang, H.; Moultos, O. A.; Economou, I. G.; Panagiotopoulos, A. Z. Gaussian-Charge Polarizable and Nonpolarizable Models for CO2. *J. Phys. Chem. B* **2016**, *120* (5), 984−994.

(23) Yu, K.; Schmidt, J. R. Many-body effects are essential in a physically motivated CO2 force field. *J. Chem. Phys.* **2012**, *136* (3), No. 034503.

(24) Zhang, Z.; Duan, Z. An optimized molecular potential for carbon dioxide. *J. Chem. Phys.* **2005**, *122* (21), No. 214507.

(25) Morales, A. D. C.; Economou, I. G.; Peters, C. J.; Siepmann, J. I. Influence of simulation protocols on the efficiency of Gibbs ensemble Monte Carlo simulations. *Mol. Simul.* **2013**, *39* (14−15), 1135−1142.

(26) Riera, M.; Yeh, E. P.; Paesani, F. Data-Driven Many-Body Models for Molecular Fluids: CO(2)/H(2)O Mixtures as a Case Study. *J. Chem. Theory Comput.* **2020**, *16* (4), 2246−2257.

(27) Mathur, R.; Muniz, M. C.; Yue, S.; Car, R.; Panagiotopoulos, A. Z. First-Principles-Based Machine Learning Models for Phase Behavior and Transport Properties of CO(2). *J. Phys. Chem. B* **2023**, *127* (20), 4562−4569.

(28) Friederich, P.; Hase, F.; Proppe, J.; Aspuru-Guzik, A. Machine-learned potentials for next-generation matter simulations. *Nat. Mater.* **2021**, *20* (6), 750−761.

(29) Meinshausen, M.; Meinshausen, N.; Hare, W.; Raper, S. C.; Frieler, K.; Knutti, R.; Frame, D. J.; Allen, M. R. Greenhouse-gas emission targets for limiting global warming to 2 degrees C. *Nature* **2009**, *458* (7242), 1158−1162.

(30) Seinfeld, J. H.; Pandis, S. N.; Noone, K. Atmospheric Chemistry and Physics: From Air Pollution to Climate Change. *Phys. Today* **1998**, *51* (10), 88−90.

(31) Falkowski, P.; Scholes, R. J.; Boyle, E.; Canadell, J.; Canfield, D.; Elser, J.; Gruber, N.; Hibbard, K.; Högberg, P.; Linder, S.; Mackenzie, F. T.; Moore, B., III; Pedersen, T.; Rosenthal, Y.; Seitzinger, S.; Smetacek, V.; Steffen, W. The Global Carbon Cycle: A Test of Our Knowledge of Earth as a System. *Science* **2000**, *290* (13), 291−296.

Downloaded from pubs.acs.org/jctce/article-pdf/21/12/5905/4126116/jctc5c00484.pdf by ZHENGZHOU UNIV user on 09 October 2026

(32) Aminu, M. D.; Nabavi, S. A.; Rochelle, C. A.; Manovic, V. A review of developments in carbon dioxide storage. *Appl. Energy* **2017**, *208*, 1389−1419.
(33) Michael, K.; Golab, A.; Shulakova, V.; Ennis-King, J.; Allinson, G.; Sharma, S.; Aiken, T. Geological storage of CO2 in saline aquifers—A review of the experience from existing storage operations. *Int. J. Greenhouse Gas Control* **2010**, *4* (4), 659−667.
(34) Aimoli, C. G.; Maginn, E. J.; Abreu, C. R. Transport properties of carbon dioxide and methane from molecular dynamics simulations. *J. Chem. Phys.* **2014**, *141* (13), No. 134101.
(35) Costentin, C.; Robert, M.; Saveant, J. M. Catalysis of the electrochemical reduction of carbon dioxide. *Chem. Soc. Rev.* **2013**, *42* (6), 2423−2436.
(36) Artz, J.; Muller, T. E.; Thenert, K.; Kleinekorte, J.; Meys, R.; Sternberg, A.; Bardow, A.; Leitner, W. Sustainable Conversion of Carbon Dioxide: An Integrated Review of Catalysis and Life Cycle Assessment. *Chem. Rev.* **2018**, *118* (2), 434−504.
(37) Velty, A.; Corma, A. Advanced zeolite and ordered mesoporous silica-based catalysts for the conversion of CO(2) to chemicals and fuels. *Chem. Soc. Rev.* **2023**, *52* (5), 1773−1946.
(38) Wu, J. Classical density functional theory for molecular systems. In *Variational Methods in Molecular Modeling*; Springer: Singapore, 2017; pp 65−99.
(39) Wu, J.; Prausnitz, J. M. *Fundamentals and Practice in Statistical Thermodynamics*; Wiley: Hoboken, NJ, 2024.
(40) Kiss, P. T.; Baranyai, A. A systematic development of a polarizable potential of water. *J. Chem. Phys.* **2013**, *138* (20), No. 204507.
(41) Buckingham, R. A. The classical equation of state of gaseous helium, neon and argon. *Proc. R. Soc. London, Ser. A* **1938**, *168* (933), 264−283.
(42) Kong, C. L.; Chakrabarty, M. R. Combining Rules for Intermolecular Potential Parameters. III. Application to the Exp 6 Potential. *J. Phys. Chem. A* **1973**, *77* (22), 2668−2670.
(43) Lu, L.; Jin, P.; Pang, G.; Zhang, Z.; Karniadakis, G. E. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nat. Mach. Intell.* **2021**, *3* (3), 218−229.
(44) Chen, T.; Chen, H. Universal approximation to nonlinear operators by neural networks with arbitrary activation functions and its application to dynamical systems. *IEEE Trans. Neural Networks* **1995**, *6* (4), 911−917.
(45) Grady, T. J.; Khan, R.; Louboutin, M.; Yin, Z.; Witte, P. A.; Chandra, R.; Hewett, R. J.; Herrmann, F. J. Model-parallel Fourier neural operators as learned surrogates for large-scale parametric PDEs. *Comput. Geosci.* **2023**, *178*, No. 105402.
(46) Yang, J.; Pan, R.; Sun, J.; Wu, J. High-Dimensional Operator Learning for Molecular Density Functional Theory. arXiv:2411.03698. arXiv.org e-Print archive. https://arxiv.org/abs/2411.03698.
(47) Shi, Y.; Lavrentiadis, G.; Asimaki, D.; Ross, Z. E.; Azizzadenesheli, K. Broadband Ground-Motion Synthesis via Generative Adversarial Neural Operators: Development and Validation. *Bull. Seismol. Soc. Am.* **2024**, *114* (4), 2151−2171.
(48) Yang, Y.; Gao, A. F.; Azizzadenesheli, K.; Clayton, R. W.; Ross, Z. E.; Yang, Y.Rapid Seismic Waveform Modeling and Inversion With Neural Operators *IEEE Trans. Geosci. Remote Sens.* 2023; Vol. *61* .
(49) Yang, Y.; Gao, A. F.; Castellanos, J. C.; Ross, Z. E.; Azizzadenesheli, K.; Clayton, R. W. Seismic Wave Propagation and Inversion with Neural Operators. *Seismic Record* **2021**, *1* (3), 126−134.
(50) Darden, T.; York, D.; Pedersen, L. Particle mesh Ewald: An N·log(N) method for Ewald sums in large systems. *J. Chem. Phys.* **1993**, *98* (12), 10089−10092.
(51) Liu, C.; Wang, J. Distilling dynamical knowledge from stochastic reaction networks. *Proc. Natl. Acad. Sci. U.S.A.* **2024**, *121* (14), No. e2317422121.
(52) Gong, X.; Louie, S. G.; Duan, W.; Xu, Y. Generalizing deep learning electronic structure calculation to the plane-wave basis. *Nat. Comput. Sci.* **2024**, *4* (10), 752−760.
(53) Chen, C.; Ong, S. P. A universal graph deep learning interatomic potential for the periodic table. *Nat. Comput. Sci.* **2022**, *2* (11), 718−728.
(54) Zhang, D.; Liu, X.; Zhang, X.; Zhang, C.; Cai, C.; Bi, H.; Du, Y.; Qin, X.; Peng, A.; Huang, J.; Li, B.; Shan, Y.; Zeng, J.; Zhang, Y.; Liu, S.; Li, Y.; Chang, J.; Wang, X.; Zhou, S.; Liu, J.; Luo, X.; Wang, Z.; Jiang, W.; Wu, J.; Yang, Y.; Yang, J.; Yang, M.; Gong, F.-Q.; Zhang, L.; Shi, M.; Dai, F.-Z.; York, D. M.; Liu, S.; Zhu, T.; Zhong, Z.; Lv, J.; Cheng, J.; Jia, W.; Chen, M.; Ke, G.; E, W.; Zhang, L.; Wang, H. DPA-2: a large atomic model as a multi-task learner. *npj Comput. Mater.* **2024**, *10* (1), No. 293.

Downloaded from pubs.acs.org/jctcce/article-pdf/21/12/5905/4126116/5c00484.pdf by ZHENGZHOU UNIV user on 09 October 2026

https://doi.org/10.1021/acs.jctc.5c00484
J. Chem. Theory Comput. 2025, 21, 5905−5915